Shaders Final Project

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Video link: https://media.oregonstate.edu/media/1_i6vjol76

Accepted Proposal

I would like to do different water wave motions for the final project, I'll make it looks like a lake surrounded by mountains. I can make the water turn into ice by stopping the wave and mix some white color into it and add reflection to it. I will not use the Gerstner Wave Equations since it is offered in the lecture PDF.

Description:

I searched for different wave equations and do the calculate of the normal. First, I add a Sinusoidal wave cause it's the most natural representation of how many things in nature change state and has been used for sound and water wave. However, I found out that Gerstner wave is also based on Sinusoidal wave equations, therefore it looks similar to each other, but since I finished it, I kept it.

The second wave equation I added is a Square Wave which is a non-sinusoidal waveform that alternates between two discrete levels (typically high and low) at regular intervals. I create the sign function when the input is positive, the output is 1 an when the input is negative, the output is -1 to make the wave looks like square.

The third wave equation I added is a sawtooth wave which rises linearly from its lowest value to its highest value and then drops suddenly back to its lowest value before repeating.

The fourth water wave I added is Stoke waves simulating shallow water, there's a parameter H in the equation represents the water depth, which is the distance from the water surface to the bottom. I set the value to 2 which is relatively small to simulate shallow water.

After adding the water equations, I added a uniform float uTemp which represent the temperature. Basically when uTemp is under 0, the water will stop, be mixed with color closed to white and have more reflection on it.

Last, I use cube mapping to make the scene looks like a beach.

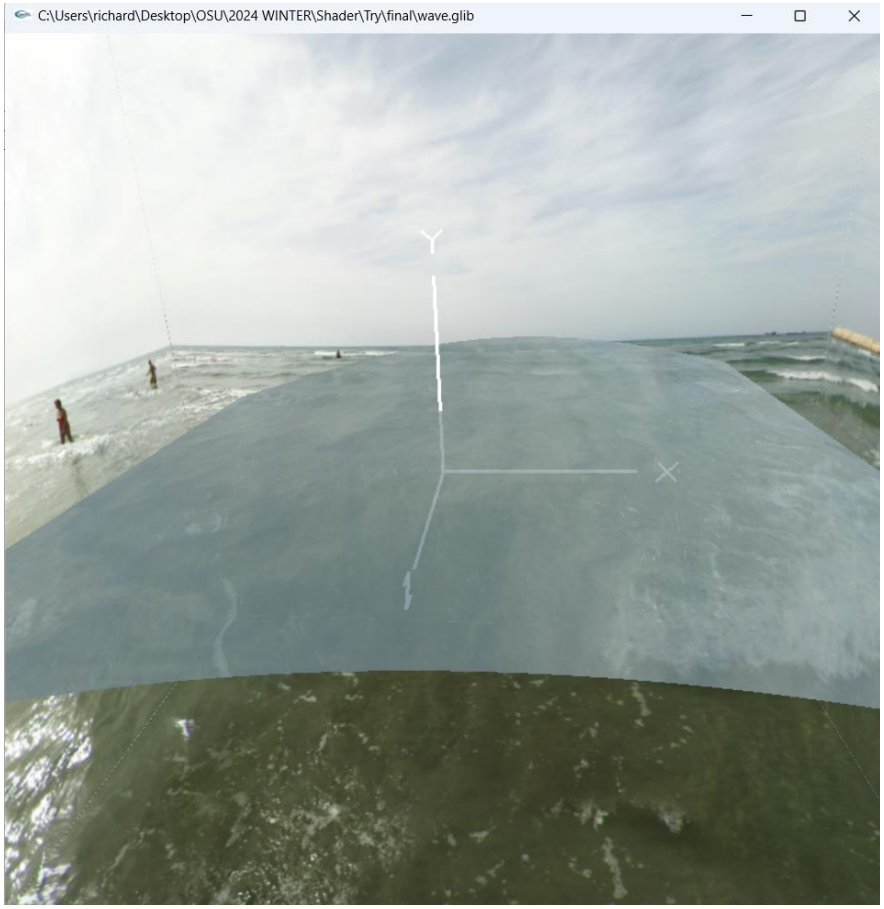

Figure 1 Sinusoidal wave

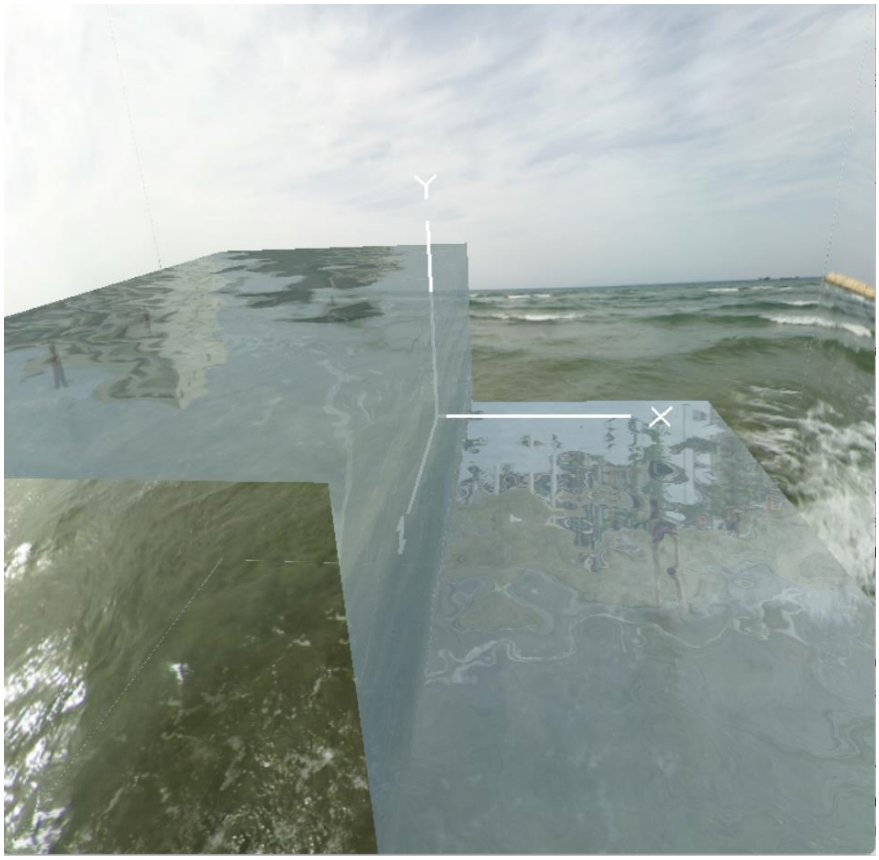

Figure 2 Square Wave

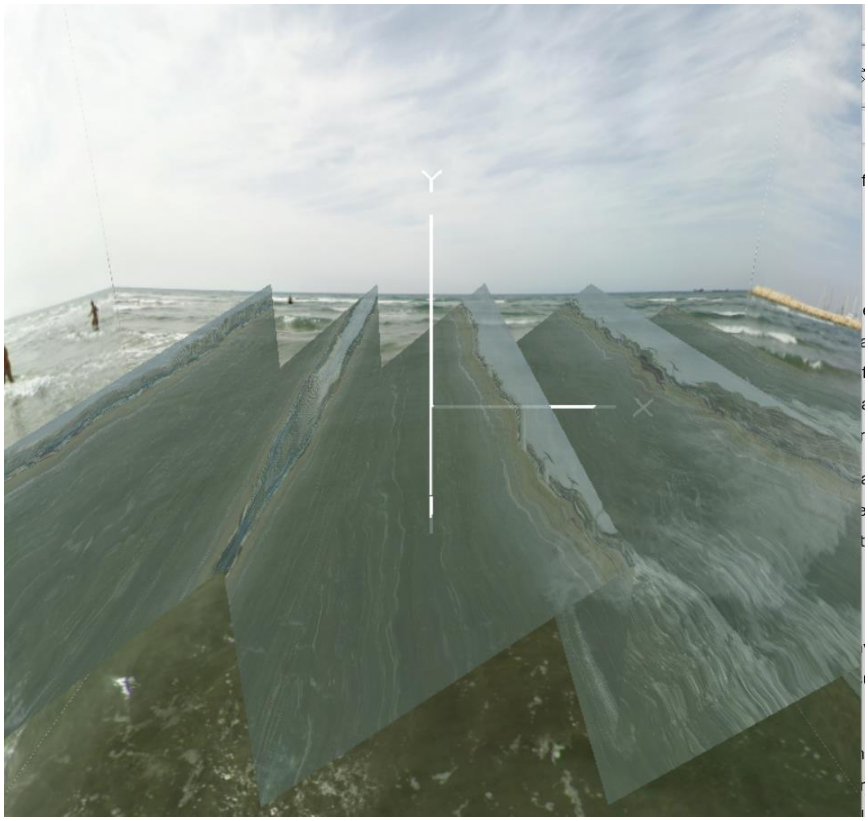

Figure 3 Sawtooth wave

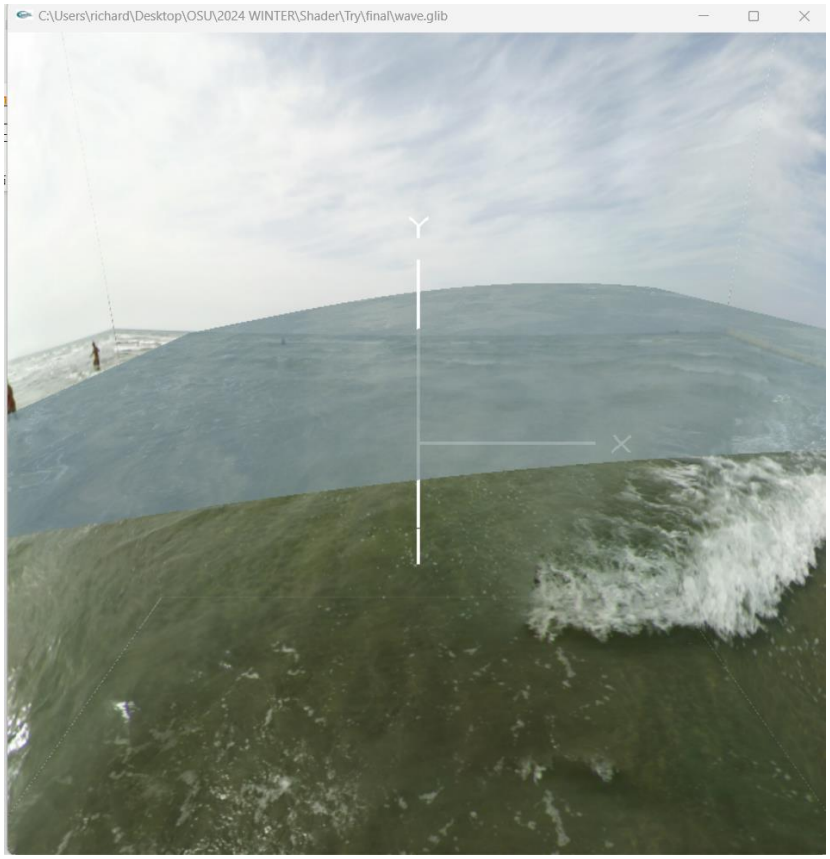

Figure 4 Stroke wave (shallow water)